

5. Correlation Heatmap

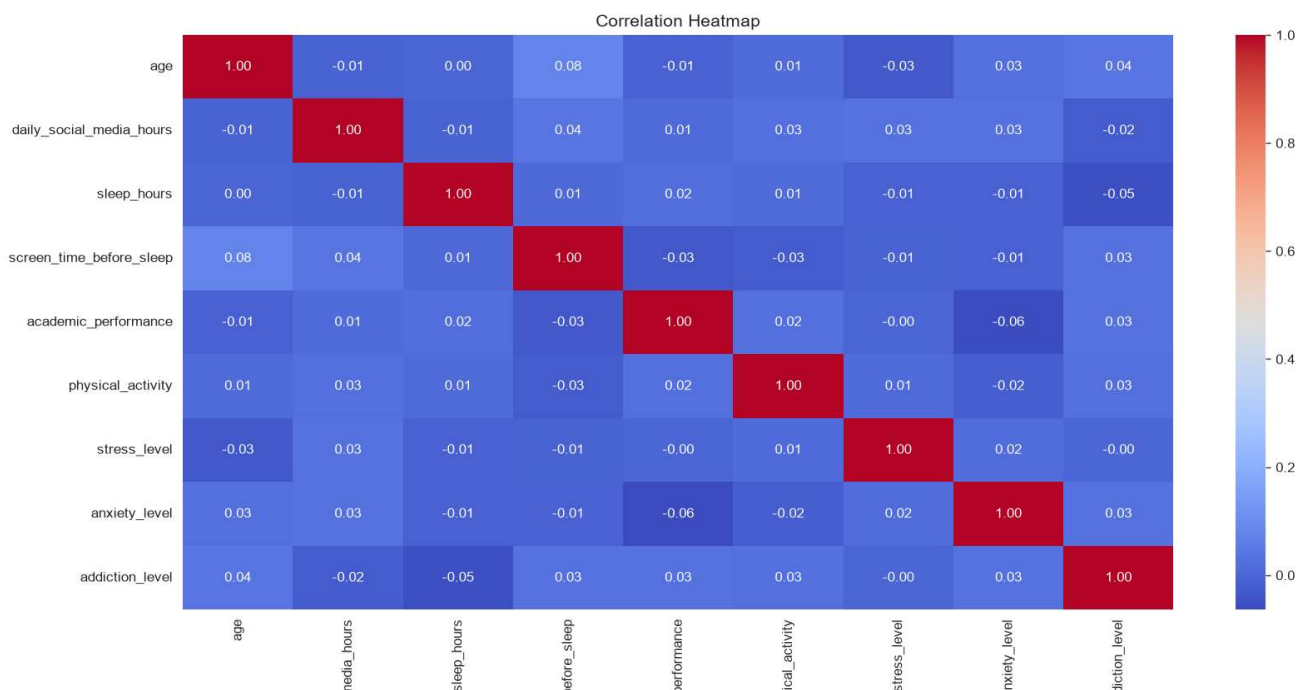
Description

The Correlation Heatmap visualizes the strength and direction of linear relationships among all numerical variables in the Teen Mental Health dataset. Correlation analysis is one of the most widely used techniques in Exploratory Data Analysis because it helps identify whether two variables increase together, decrease together, or exhibit little measurable association.

Python Code

```
plt.figure(figsize=(10,8))
sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,cmap='coolwarm',fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```

Graph



Interpretation

The heatmap indicates that most correlation coefficients are close to zero, suggesting weak linear relationships among the variables included in this dataset. This implies that no single variable strongly explains mental health outcomes, and psychological well-being is likely influenced by multiple interacting behavioural and environmental factors. Although strong correlations are not observed, the heatmap remains an essential exploratory tool because it helps identify variables requiring further investigation. It also demonstrates that more advanced analytical methods, such as regression modelling or machine learning algorithms, may be necessary to uncover complex patterns that cannot be detected through simple correlation analysis.